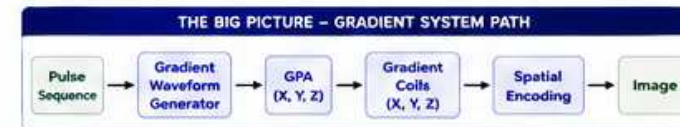


Gradient problems leave characteristic image patterns. Recognize → Test → Isolate → Fix → Verify

Recognize (Image Clues) → Isolate (Tests) → **Fix (Root Cause)** → **Verify (Image & Metrics)**



1. COMMON GRADIENT FAILURE PATTERNS		
Pattern (What You See)	Typical Appearance	Likely Gradient Issue
Geometric Distortion / Warping		Nonlinear gradient, eddy currents, gradient delay, wrong FOV prescale/POST
N/2 Ghosting (EPI)		EPI phase-encode issues: mismatch, timing, eddy currents, polarity
Image Shift		Wrong gradient offset, FOV/center error, gradient delay, eddy currents
FOV/Scale Error (Too Big/Small)		Mis-calibrated gradients, wrong FOV, wrong frequency/phase scaling
Directional Ghosting (Blurring/Repeats)		Faulty phase-encode gradient, timing, spoiling, or cables
Zipper / Streaking / Ringing		Eddy currents, gradient delay, inadequate dead time, high dB/dt
Vibration / Acoustic		Loose gradient mount, cable clamp, cryo piping, room vibration
Axis-Specific Failure		Problem follows orientation of the faulty axis

2. HOW TO USE IMAGE CLUES	
Change something and watch what happens. This tells you what system is responsible.	
CHANGE & OBSERVE PRINCIPLES	
↔ Swap Frequency ↔ Phase If artifact changes → likely encoding related (gradients/eddy currents). If same → not encoding.	
🔄 Rotate / Re-orient the Scan If artifact rotates with the image → gradient related. If stays in patient → anatomy / motion.	
↔ Reverse Phase Encode (AP ↔ PA) If artifact reverses → gradient / timing related.	
📶 Change Bandwidth / Echo Spacing / TE If artifact changes → gradient / timing / eddy. If no change → other cause.	
☰ Use Different Sequence If artifact appears across many sequences → hardware. If only one → sequence/parameter.	
QUICK RULE OF THUMB	
- Follows the anatomy = likely patient / coil / motion - Follows the encoding direction = likely gradient - Follows physical axis (X/Y/Z) = axis or hardware problem - Present everywhere, even in prescan = system level	

3. ISOLATE THE PROBLEM (KEY TESTS)		
Test	What it Checks	How to Do It
Prescan / Coil Check (no patient)	System baseline (no patient effects)	Run prescan. Look for distortions, shifts, spikes, noise.
Geometry / FOV Test	FOV accuracy, linear gradients	Acquire grid phantom. Check geometry.
Gradient Impulse Test	Gradient performance, amplitude, delays	Service Browser → Gradient Impulse.
Eddy Current Test	Eddy currents & settling	Service Browser → Eddy Current Measurement.
PNS / dB/dt Monitor	High dB/dt, ramps, acoustic vibration	Check PNS & dB/dt logs during scan.
B0 Field Map	B0 homogeneity (not gradient but confounds distortion)	Run field map. If B0 poor → fix shimming.
Orientation Test	Axis isolation	Rotate patient (sag/cor/tra). See if artifact follows axis.
Reconstruction Test	Ringing / truncation	Change matrix, FOV, filter, zero-fill.

4. COMMON ROOT CAUSES & WHERE TO LOOK		
Root Cause	Symptoms / Clues	Where to Inspect / Check
Gradient Coil / Axis Problem (Open, short, or mismatch)	Axis-specific artifact, noise, heating	Coil resistance, connectors, Service Browser (Impedance)
GPA (X/Y/Z) Fault	Loss of power, artifacts, PNS trips, errors	GPA status, logs, faults, heatsink temp
Loose / Damaged Cables or Connectors	Intermittent artifacts, spikes, axis-specific	Gradient cabling, amp-to-coil connections
Eddy Currents	Zipper, ghosts, blurring on EPI	Eddy current test, gradient delay, shielding
Gradient Delay	Geometric distortion, N/2 ghosting	Gradient delay calibration, service tools
Wrong Offsets / Gains	Image shift, scaling errors	Gradient offsets, FOV, center, prescale
High dB/dt / PNS Limits	PNS warnings, scan aborts, ringing	Reduce dB/dt, ramp times, check PNS settings
Vibration / Mechanical	Ringing, noise, image breakup	Tighten mounts, cable clamps, cryo lines
Gradient Nonlinearity	Edge warping, nonlinear geometry	Linearize gradients, update calibration (GEOMETRY)

5. BEFORE & AFTER EXAMPLES (REAL-WORLD PATTERNS)					
Problem (What You See)	After Fix (Expected Result)	N/2 Ghosting (EPI) (Other fix)	Image Shift	Eddy Current (Streaking)	Scaling / FOV Error
Fixed delay / linearization		Adjusted phase encode timing / polarity	Corrected gradient offset	Improved eddy current correction	Corrected FOV / prescale
					Axis-Specific Failure (Sagittal)
					Repaired Y axis / cable

6. ORIENTATION TEST EXAMPLE	
Same Artifact – Different Orientations	
Sagittal	Coronal
Axial	
Artifact bands run left-right in the image in all 3 planes → Corblem is in the phase-encode (Y) axis.	

7. FREQUENCY / PHASE SWAP EXAMPLE	
Original (Phase AP)	Phase Reversed (Phase PA)
N/2 ghosting reverses direction when phase encoding is reversed → Confirms phase-encode / timing issue.	

8. GRADIENT SYSTEM HEALTH CHECKLIST	
- Prescan clean (no patient) - No unusual noise / vibration - PNS / dB/dt within safe limits - Gradient impulse test within spec - Eddy current test within spec	- Geometry within tolerance - No image shift - No N/2 ghosts on EPI - No distortion on long FOV - Logs show no gradient faults

9. SERVICE TOOLS (GE EXAMPLES)	
Tool / Test	Purpose
Service Browser → Gradient Impulse	Check amplitude, rise time, ring
Service Browser → Eddy Current Measurement	Measure eddy currents & decay
Service Browser → Gradient Impedance	Check coil continuity / balance
Service Browser → PNS Monitor	Monitor dB/dt, ensure within limits
Service Browser → Gradient Delay Test	Calibrate gradient timing
Service Browser → Geometry (FOV) Tool	Verify linearity & accuracy

10. KEY TAKEAWAYS	
🔍	Gradient problems have distinctive image signatures.
⚡	Use simple changes to isolate the cause.
📶	Test systematically: prescan → impulse → eddy → orientation.
🔧	Fix the root cause, not just the symptom.
📄	Always verify with images and metrics.

11. QUICK REFERENCE FLOW	
1	See artifact → Recognize pattern
2	Prescan (no patient) → Is it system?
3	Swap Freq/Phase or Re-orient → Does it change?
4	Run Tests (Impulse, Eddy, Geometry, Logs)
5	Isolate cause (axis / cable / GPA / settings)
6	Fix root cause (repair, calibrate, adjust)
7	Verify → Images + Metrics → Document



ENGINEER NOTES

- Always document changes and results.
- Compare to baseline or another identical system.
- If in doubt, escalate with data.

ABBREVIATIONS

FOV = Field of View	dB/dt = Rate of Change of Gradient Field
GPA = Gradient Power Amplifier	TE = Echo Time
PNS = Peripheral Nerve Stimulation	AP/PA = Anterior-Posterior / Posterior-Anterior
EPI = Echo Planar Imaging	



SAFETY REMINDER

- High dB/dt can cause PNS or patient discomfort.
- Follow all site safety procedures.
- Verify gradient cooling and ventilations.



THE BOTTOM LINE

See it. Understand it. Test it. Fix it. Verify it.
Great gradients = accurate images = great diagnosis.



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